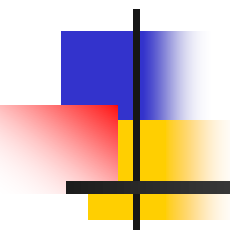
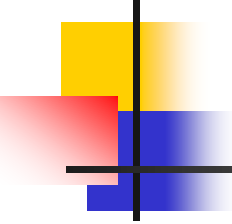


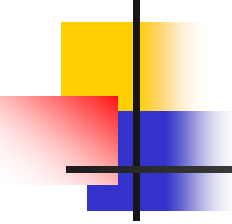
Plan Generation & Causal-Link Planning 1





Review

- Frame problem:
 - How to specify what does not change
- Qualification problem
 - Hard to specify all preconditions
- Ramification problem
 - Hard to specify all effects



Review

- Situation Calculus

- Successor state axioms:

$$\begin{aligned} \text{Broken}(x, \text{do}(s, a)) \leftrightarrow & [a = \text{drop}(x) \wedge \text{fragile}(x, s)] \vee \\ & \exists b[a = \text{explode}(b) \wedge \text{nextTo}(b, x, s)] \vee \\ & \text{broken}(x, s) \wedge \neg \exists a = \text{repair}(x) \end{aligned}$$

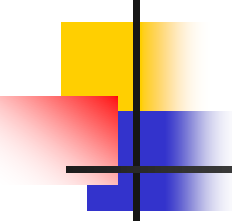
- Preconditions axioms:

$$\begin{aligned} \text{Poss}(\text{pickup}(r, x), s) \leftrightarrow & \text{robot}(r) \wedge \forall z \neg \text{holding}(z, x, s) \wedge \\ & \text{nextTo}(r, x, s) \end{aligned}$$

- Strips representation

- Means-ends analysis

- Networks of Actions (Noah)

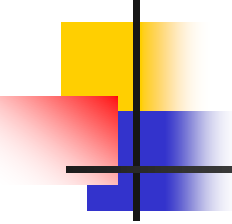


Domain-Independent Planning

Inputs:

- Domain Action Theory
- Problem Instance
 - Description of (initial) state of the world
 - Specification of desired goal behavior

Output: sequence of actions that
executed in initial state satisfy goal

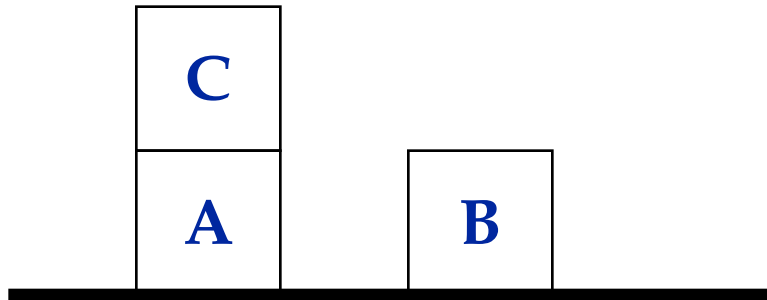


“Classical Planning” Assumptions

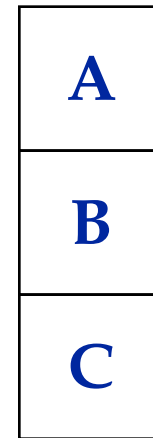
- Atomic Time
 - Instantaneous actions
- Deterministic Effects
- Omniscience
- Sole agent of change
- Goals of attainment

Example Problem Instance: “Sussman Anomaly”

Initial State:



Goal:

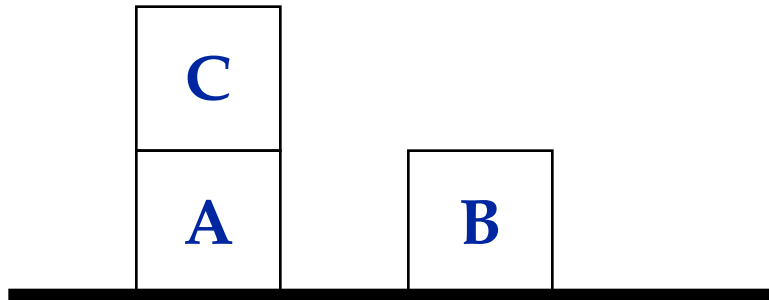


Initial State: (on-table A) (on C A) (on-table B)
(clear B) (clear C)

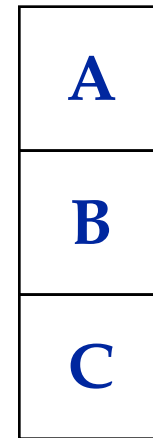
Goal: (on A B) (on B C)

Example Problem Instance: “Sussman Anomaly”

Initial State:

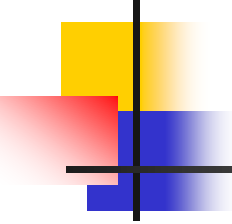


Goal:



Initial State: (and (on-table A) (on C A)
(on-table B) (clear B) (clear C))

Goal: (and (on A B) (on B C))



Action Representation: Propositional STRIPS

Move-C-from-A-to-Table:

preconditions: (on C A) (clear C)

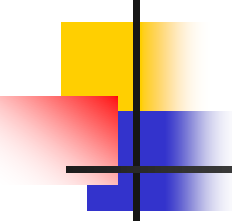
effects:

add (on-table C)

delete (on C A)

add (clear A)

Solution to frame problem: explicit effects are the only changes to the state.



Action Representation: Propositional STRIPS

Move-C-from-A-to-Table:

preconditions: (and (on C A) (clear C))

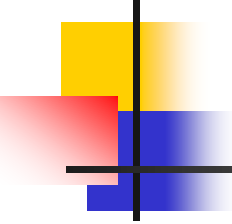
effects:

(and (on-table C)

(not (on C A))

(clear A))

Solution to frame problem: explicit effects are the only changes to the state.



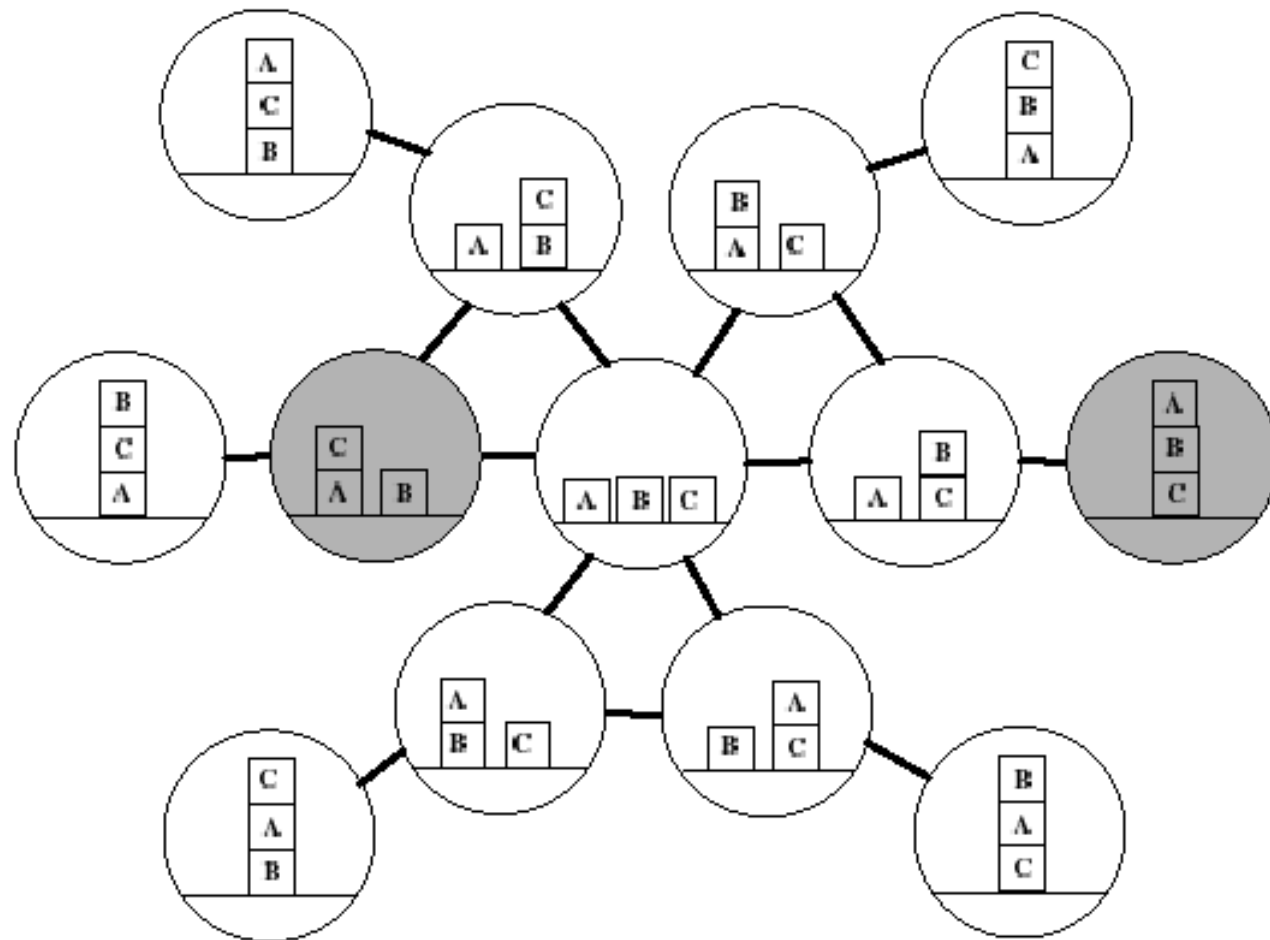
Plan Generation:

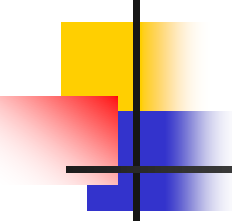
Search space of world states

Planning as a (graph) search problem

- Nodes: world states
- Arcs: actions
- Solution: path from the initial state to one state that satisfies the goal
 - Initial state is fully specified
 - There are many goal states

Search Space: Blocks World





Progression:

Forward search ($I \rightarrow G$)

ProgWS(state, goals, actions, path)

If state satisfies goals, then return path

else $a = \textit{choose}(\text{actions})$, s.t.

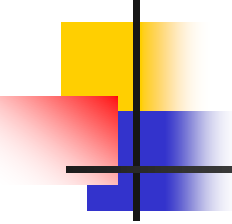
 preconditions(a) satisfied in state

if no such a , then return failure

else return

ProgWS(apply(a , state), goals, actions,
 concatenate(path, a))

First call: ProgWS(IS , G , $Actions$, $()$)



Progression Example: Sussman Anomaly

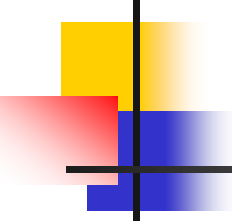
I: (on-table A) (on C A) (on-table B) (clear B) (clear C)

G: (on A B) (on B C)

- P(I, G, BlocksWorldActions, ())
- P(S1, G, BWA, (move-C-from-A-to-table))
- P(S2, G, BWA, (move-C-from-A-to-table,
move-B-from-table-to-C))
- P(S3, G, BWA, (move-C-from-A-to-table,
move-B-from-table-to-C,
move-A-from-table-to-B))

**Non-Deterministic
Choice!**

$G \subseteq S3 \Rightarrow \text{Success!}$



Regression:

Backward Search ($I \leftarrow G$)

RegWS(init-state, current-goals, actions, path)

If init-state satisfies current-goals, then return path

else $a = \textit{choose}$ (actions), s.t. some effect of a
satisfies one of current-goals

If no such a , then return failure [unachievable*]

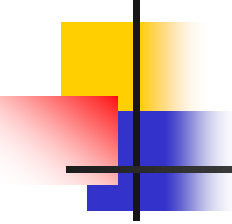
If some effect of a contradicts some of current-goals,
then return failure [inconsistent state]

$CG' = \text{current-goals} - \text{effects}(a) + \text{preconditions}(a)$

If $\text{current-goals} \subset CG'$, then return failure [useless*]

RegWS(init-state, CG' , actions, concatenate(a , path))

First call: **RegWS**(IS, G, Actions, ())



Regression Example: Sussman Anomaly

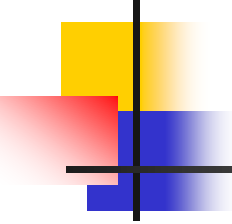
I: (on-table A) (on C A) (on-table B) (clear B) (clear C)

G: (on A B) (on B C)

**Non-Deterministic
Choice!**

- R(I, G, BlocksWorldActions, ())
- R(I, ((clear A) (on-table A) (clear B) (on B C)), BWA, (move-A-from-table-to-B))
- R(I, ((clear A) (on-table A) (clear B) (clear C), (on-table B)), BWA, (move-B-from-table-to-C, move-A-from-table-to-B))
- R(I, ((on-table A) (clear B) (clear C) (on-table B) (on C A)), BWA, (move-C-from-A-to-table, move-B-from-table-to-C, move-A-from-table-to-B))

current-goals $\subseteq$ I $\Rightarrow$ Success!



Regression Example: Sussman Anomaly

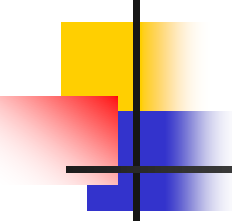
I: (on-table A) (on C A) (on-table B) (clear B) (clear C)

G: (on A B) (on B C)

- P(I, G, BlocksWorldActions, ())
- P(S1, G, BWA, (move-C-from-A-to-table))
- P(S2, G, BWA, (move-C-from-A-to-table,
move-B-from-table-to-C))
- P(S3, G, BWA, (move-C-from-A-to-table,
move-B-from-table-to-C,
move-A-from-table-to-B))

**Non-Deterministic
Choice!**

$G \subseteq S3 \Rightarrow \text{Success!}$

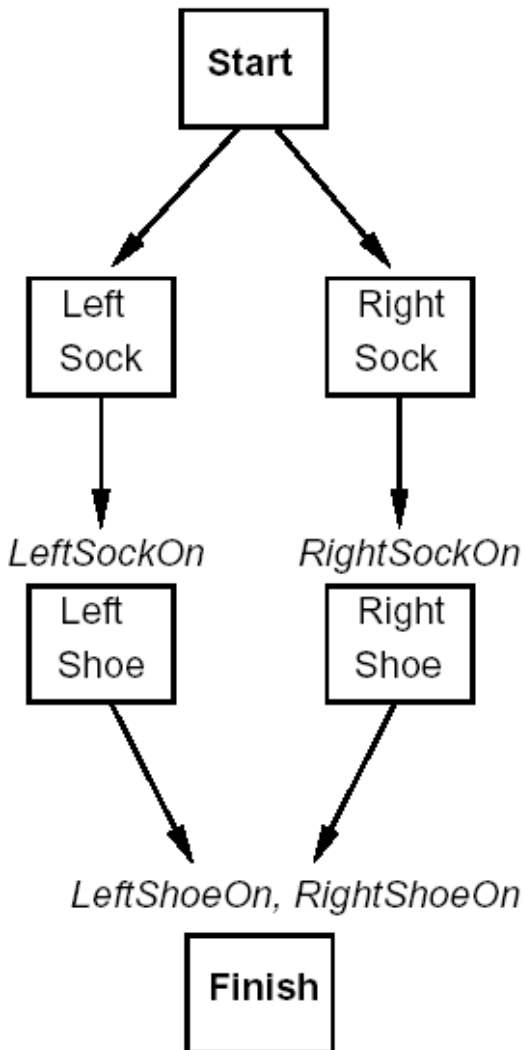


Progression vs. Regression

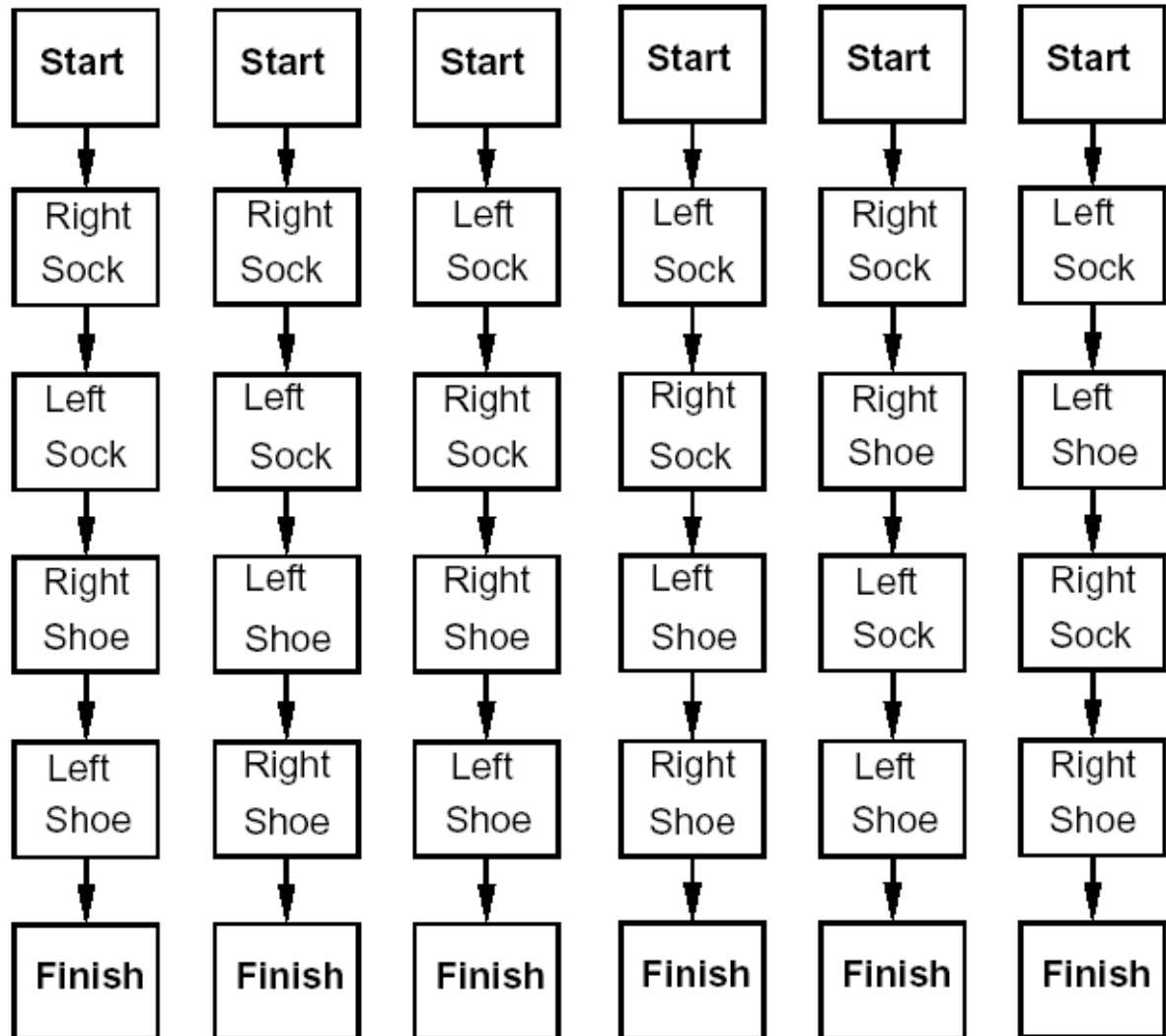
- Both algorithms are:
 - Sound: the result plan is valid
 - Complete: if valid plan exists, they find one
- Non-deterministic choice $\Rightarrow$ search!
 - Brute force: DFS, BFS, Iterative Deepening, ...
 - Heuristic: A^* , IDA^* , ...
- Complexity: $O(b^n)$ worst-case
b = branching factor, n = |“choose”|
- Regression: often smaller b, focused by goals
- Progression: full state to compute heuristics

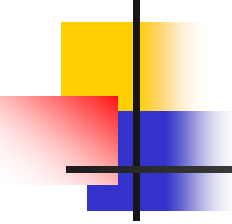
Total-Order vs Partial-Order Plans

Partial Order Plan:



Total Order Plans:





Plan Generation:

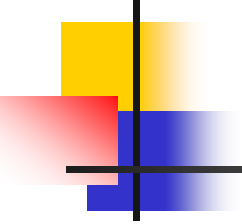
Search space of plans

Partial-Order Planning (POP)

- Nodes are partial plans
- Arcs/Transitions are plan refinements
- Solution is a node (not a path).

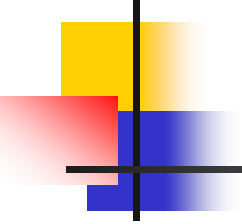
Principle of “Least commitment”

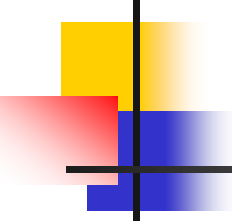
- e.g. do not commit to an order of actions until it is required



Terminology

- Step: a step in the partial plan which is bound to a specific action
- Ordering: $s1 < s2$ $s1$ must precede $s2$
- Open conditions: preconditions of the steps (including goal step)
- Causal link
- A commitment that the condition at Ac will be made true by Ap
 - Requires Ap to cause Q
 - Should have an effect Q

- 
-
- Causal link
 - Unsafe link



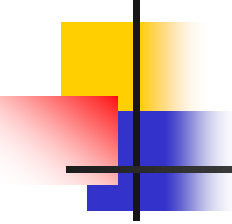
Partial Plan Representation

- Plan = (A, O, L), where
 - A: set of actions in the plan
 - O: *temporal orderings* between actions ($a < b$)
 - L: *causal links* linking actions via a literal

- Causal Link: $A_p \xrightarrow{Q} A_c$

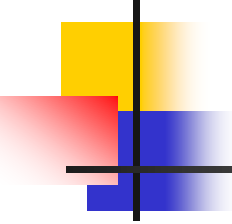
Action A_c (consumer) has precondition Q that is established in the plan by A_p (producer).

move-a-from-b-to-table $\xrightarrow{\text{(clear b)}}$ move-c-from-d-to-b



Partial plan representation

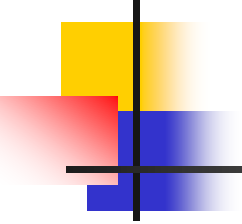
- $P = (A, O, L, OC, UL)$
- OC: set of open conditions
- UL: set of unsafe links where P is deleted by some S_k
- Flaw: Open condition and unsafe link

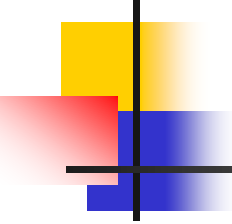


Partial Plan Representation

- Example
- Flaw:
- Solution plan: A partial plan with no remaining flaw. Every open condition must be satisfied by some action. No unsafe links should exist (i.e the plan is consistent)

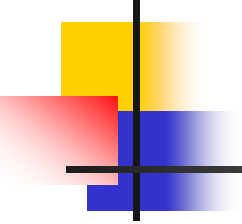
Algorithm (POP)

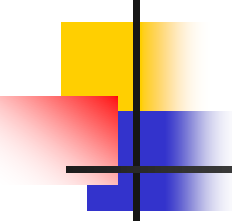
- 
1. Let P be an initial plan
 2. Flaw selection: choose a flaw f (either open condition or unsafe link)
 3. Flw resolution:
 1. If f is an open condition
 2. choose an action S that achieves f
 3. If f is an unsafe link
 1. choose promotion or demotio
 4. Update P
 5. Return NULL if no resolution exist
 6. If there is no flaw left, return P
 7. else go to 2



Example

- Initial plan
- Intermediate partial plan

- 
-
- For flaw selection we have several choices
 - Flaw selection (open condition, unsafe link)
 - Flaw resolution (how to select partial plan?)
 - Action selection (backtrack point)
 - Unsafe link selection (backtrack point)



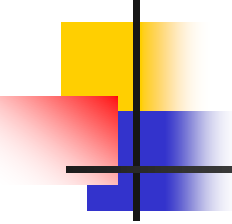
Threats to causal links

Step A_t threatens link (A_p, Q, A_c) if:

1. A_t has (not Q) as an effect, and
2. A_t could come between A_p and A_c , i.e.

$O \cup (A_p < A_t < A_c)$ is consistent

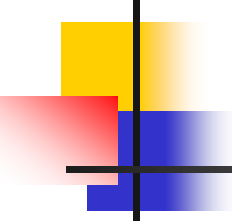
What's an example of an action that threatens the link example from the last slide?



Initial Plan

For uniformity, represent initial state and goal with two special actions:

- A_0 :
 - no preconditions,
 - initial state as **effects**,
 - must be the **first** step in the plan.
- A_∞ :
 - no effects
 - goals as **preconditions**
 - must be the **last** step in the plan.



POP algorithm

POP((A, O, L), agenda, actions)

If agenda = () then return (A, O, L)

Pick (Q, a_{need}) from agenda

a_{add} = **choose**(actions) s.t. $Q \in \text{effects}(a_{\text{add}})$

If no such action a_{add} exists, **fail**.

$L' := L \cup (a_{\text{add}}, Q, a_{\text{need}})$; $O' := O \cup (a_{\text{add}} < a_{\text{need}})$

agenda' := agenda - (Q, a_{need})

If a_{add} is new, then $A := A \cup a_{\text{add}}$ and

$\forall P \in \text{preconditions}(a_{\text{add}})$, add (P, a_{add}) to agenda'

For every action a_t that threatens any causal link (a_p, Q, a_c) in L'

choose to add $a_t < a_p$ or $a_c < a_t$ to O.

If neither choice is consistent, **fail**.

POP((A', O', L'), agenda, actions)

Termination

Goal Selection

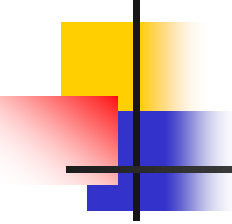
Action Selection

Update goals

Protect causal links

- Demotion: $a_t < a_p$

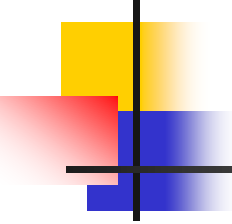
- Promotion: $a_c < a_t$



POP

POP is sound and complete

- POP Plan is a solution if:
 - All preconditions are supported (by causal links), i.e., no open conditions.
 - No threats
 - Consistent temporal ordering
- By construction, the POP algorithm reaches a solution plan

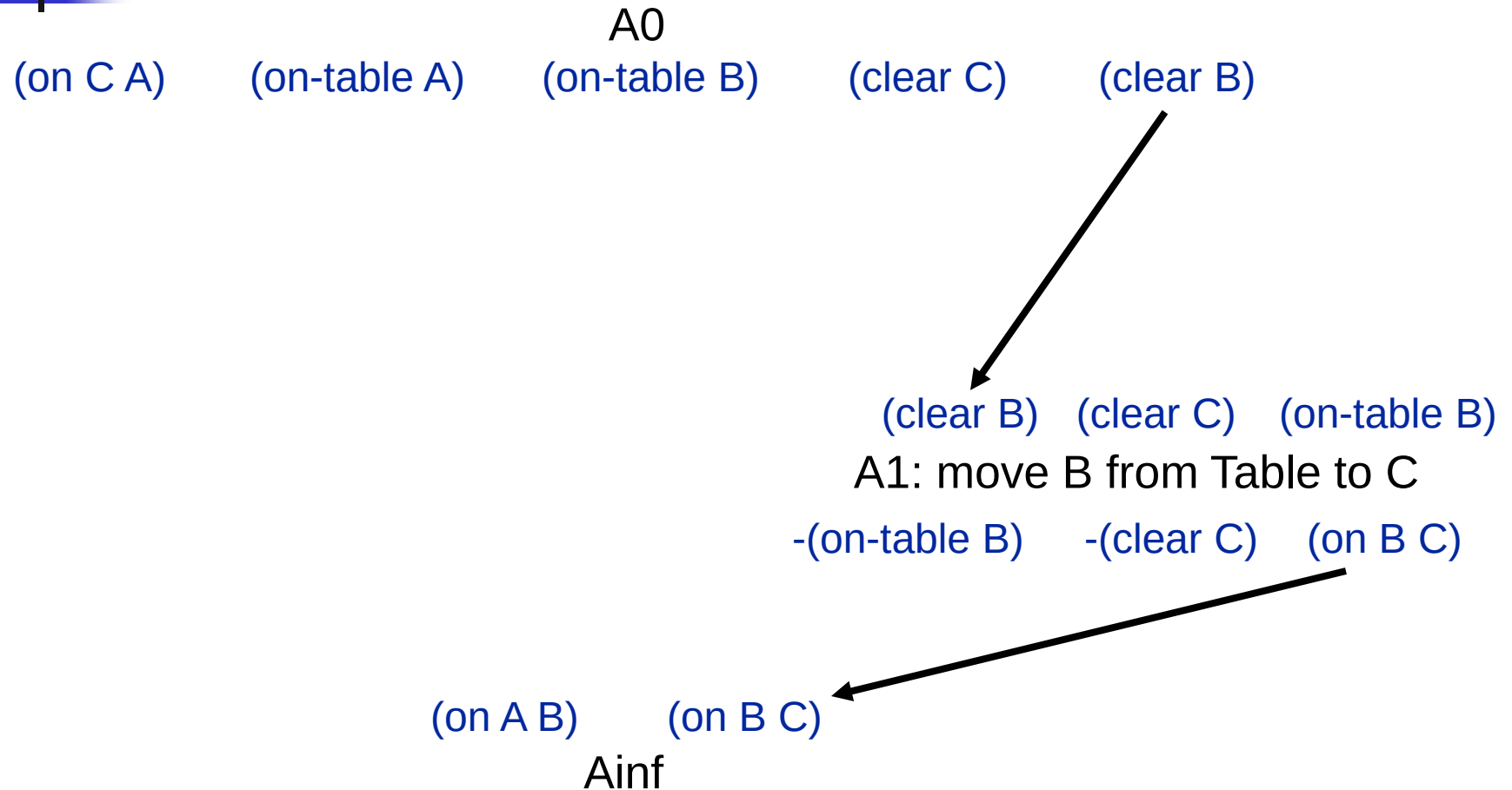


POP example: Sussman Anomaly

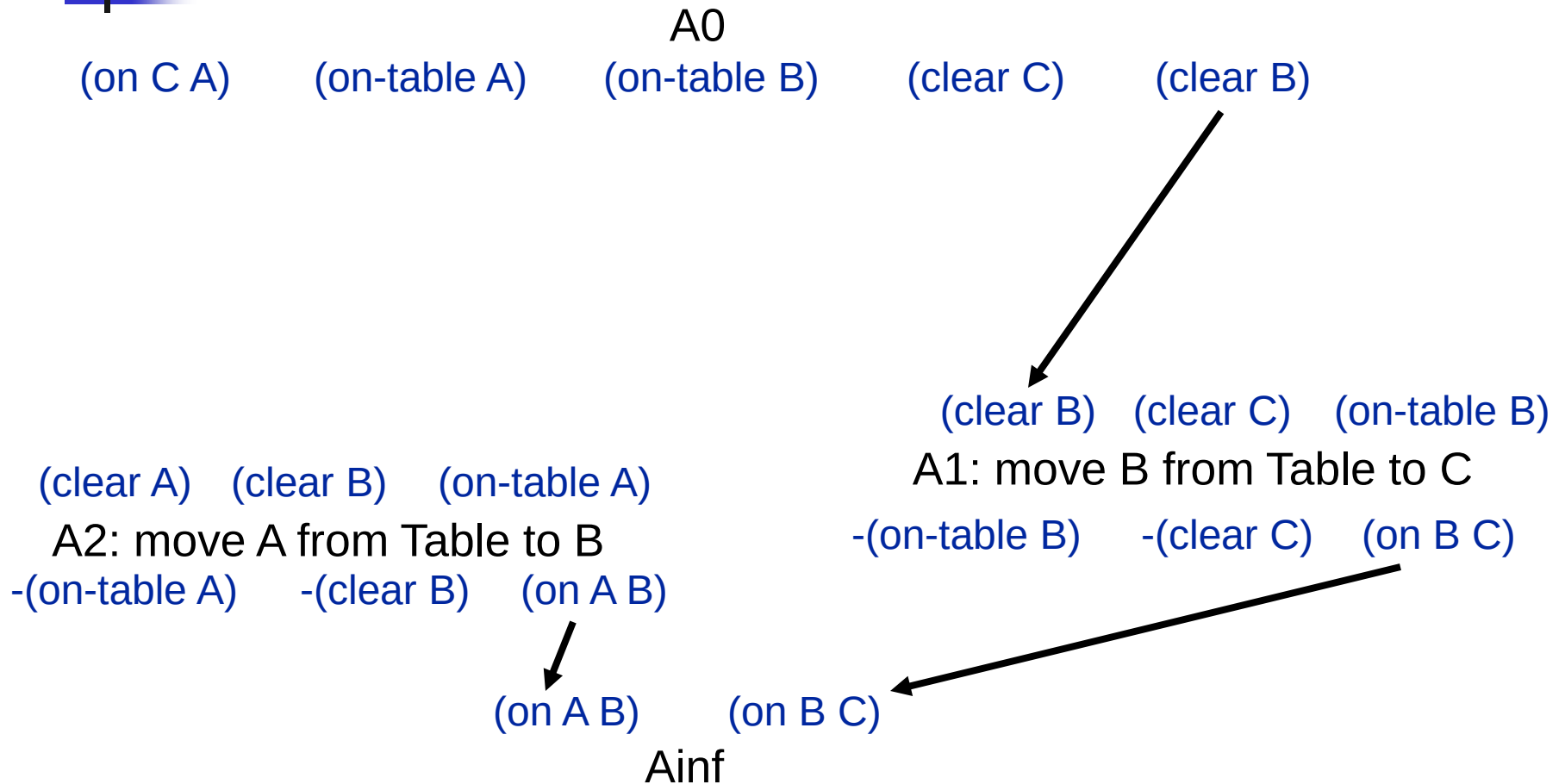
A0
(on C A) (on-table A) (on-table B) (clear C) (clear B)

(on A B) (on B C)
Ainf

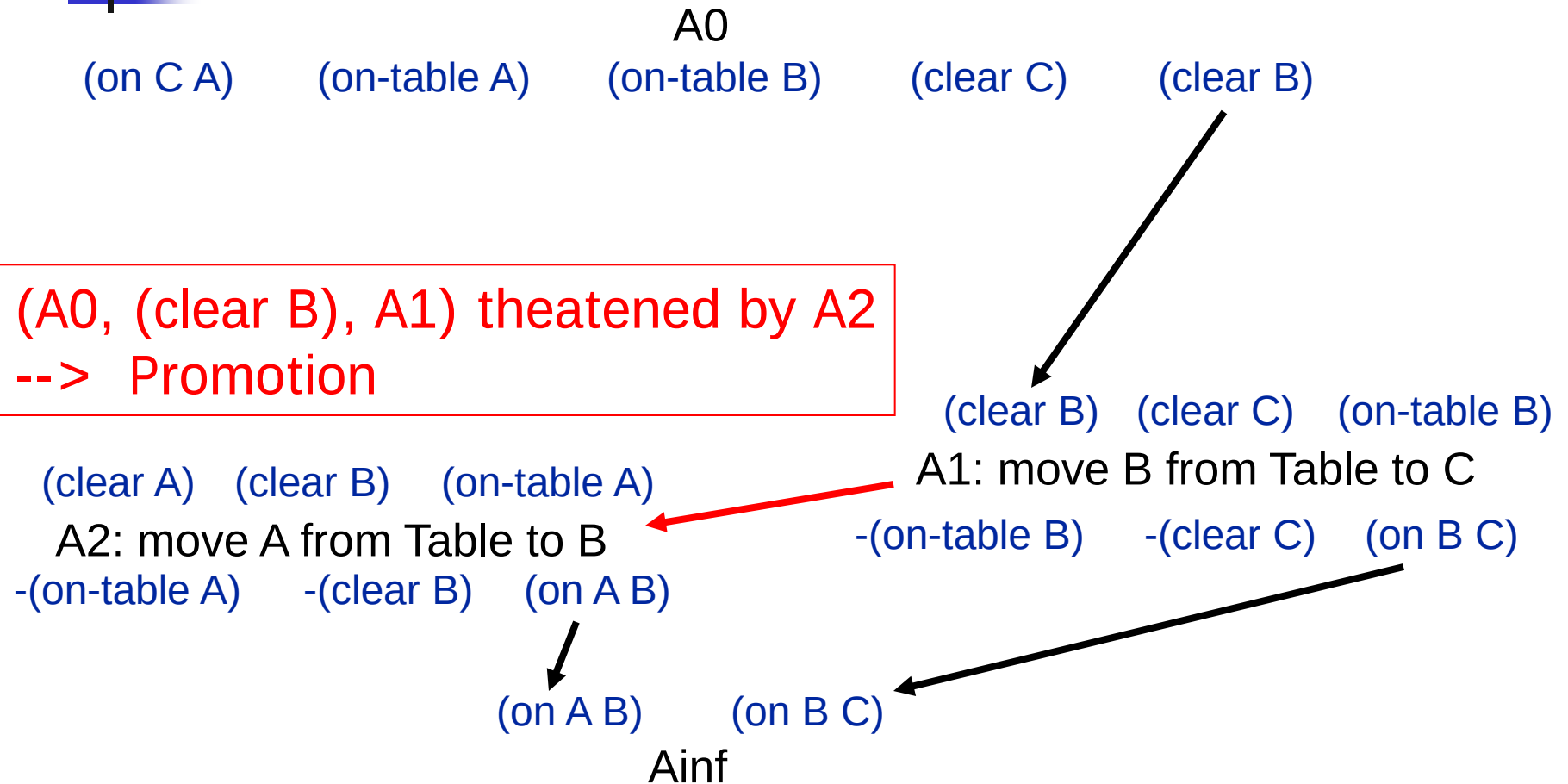
Work on open precondition (on B C) and (clear B)



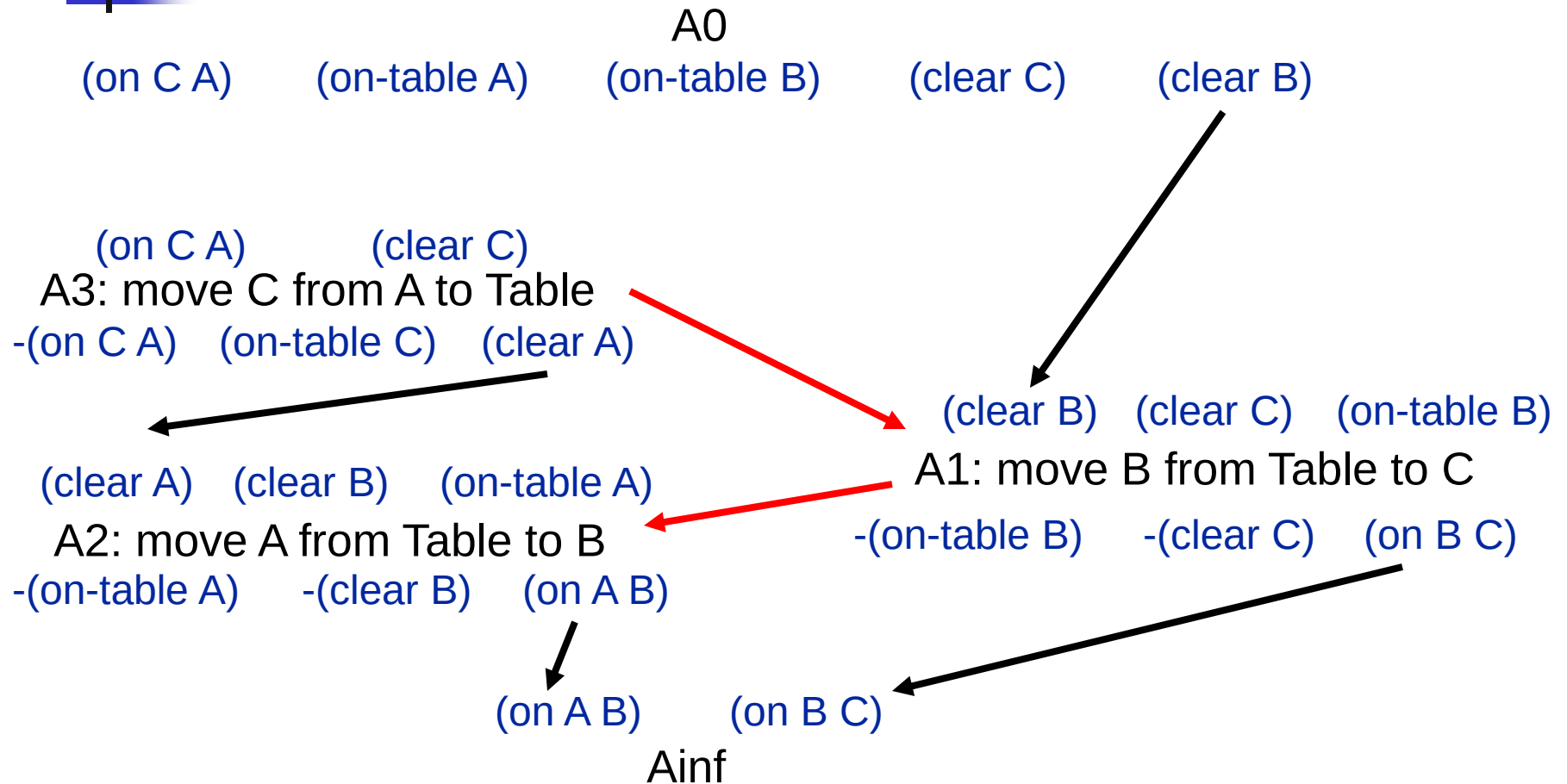
Work on open precondition (on A B)



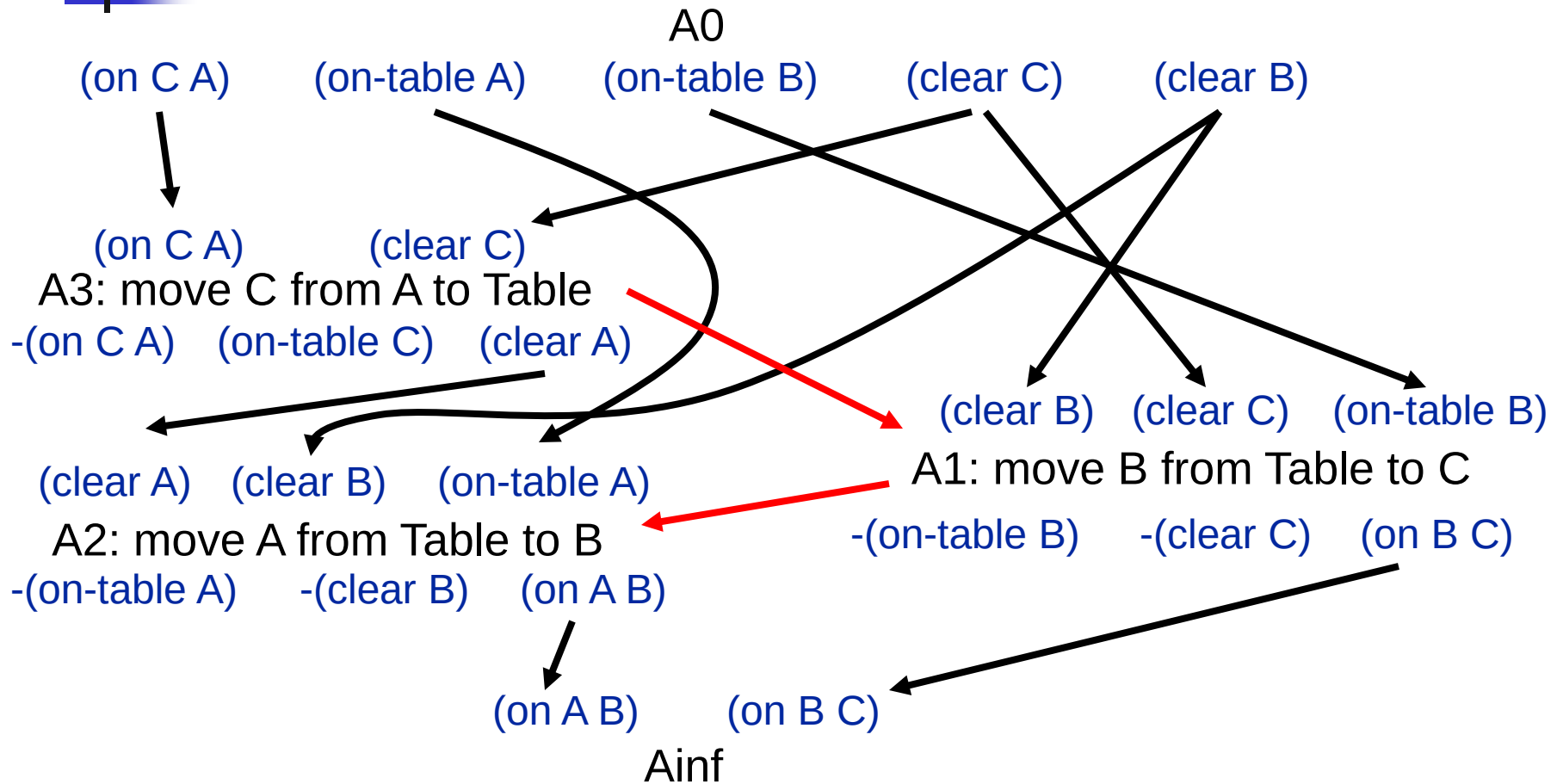
Protect causal links

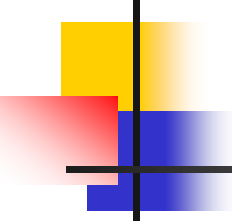


Work on open precondition (clear A) and protect links



Final plan





Partial-Order Planning vs State-Space Planning

Complexity: $O(b^n)$ worst-case

- Non-deterministic choices (n):
 - ProgWS, RegWS: $n = |\text{actions}|$
 - POP: $n = |\text{preconditions}| + |\text{link protection}|$
 - Generally an action has several preconditions
- Branching factor (b)
POP has smaller b:
 - No backtrack due to goal ordering
 - Least commitment: no premature step ordering
 - Does POP make the least possible amount of commitment?